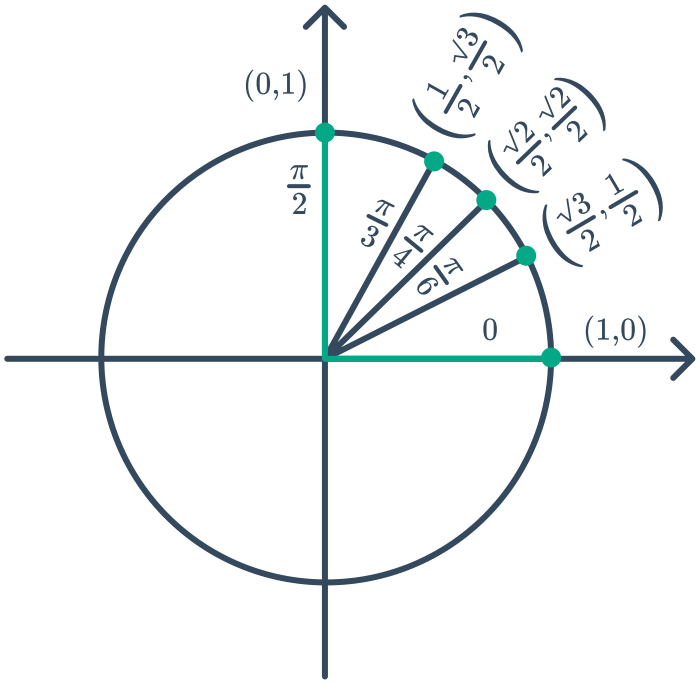
Trigonometric functions on the unit circle

1 Consider the following diagram of the unit circle:



Find the exact value of the following:

a $\cos\frac{\pi}{3}$ **b** $\sin\frac{\pi}{3}$ **c** $\sin\frac{\pi}{4}$ **d** $\cos\frac{\pi}{6}$

e $\cos\frac{\pi}{2}$ **f** $\sin\frac{\pi}{2}$ **g** $\cos 2\pi$ **h** $\tan\frac{\pi}{6}$

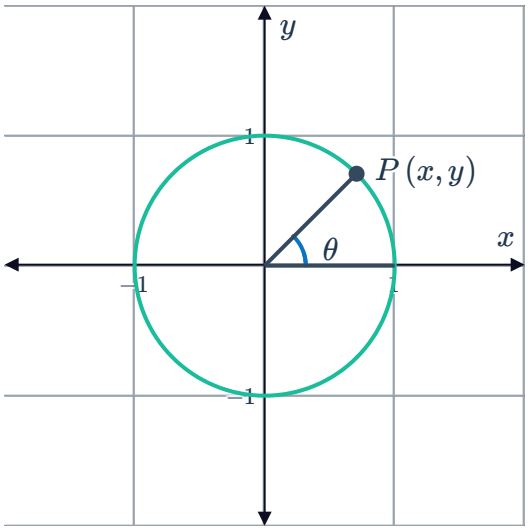
2 Find the exact values of the following:

a $\sin\left(\frac{7\pi}{6}\right)$ **b** $\cos\left(\frac{2\pi}{3}\right)$ **c** $\tan\left(\frac{7\pi}{4}\right)$ **d** $\sin\left(\frac{19\pi}{6}\right)$

e $\cos\left(-\frac{\pi}{6}\right)$ **f** $\tan\left(\frac{11\pi}{4}\right)$ **g** $\tan\left(-\frac{\pi}{6}\right)$ **h** $\sin\left(-\frac{5\pi}{6}\right)$

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3 The point P on the unit circle has the coordinates (x, y).
Write the following in terms of x and/or y:



a $\csc\theta$ **b** $\sec\theta$

c $\cot\theta$

4 Find the exact value of the following:

a $\sec(-30)^\circ$ **b** $\cot 225^\circ$ **c** $\csc 240^\circ$ **d** $\sec 300^\circ$

e $\cot 585^\circ$ **f** $\csc 945^\circ$ **g** $\sec 945^\circ$ **h** $\cot 945^\circ$

5 Find the exact value of the following:

a $\sec\frac{\pi}{6}$ **b** $\csc\frac{\pi}{3}$ **c** $\sec\left(\frac{2\pi}{3}\right)$ **d** $\csc\left(\frac{4\pi}{3}\right)$

e $\cot\left(\frac{7\pi}{6}\right)$ **f** $\sec\left(\frac{15\pi}{4}\right)$ **g** $\csc\left(-\frac{2\pi}{3}\right)$ **h** $\cot\left(\frac{25\pi}{6}\right)$

6 For each of the following functions, state:

i The quadrants where the function is positive.

ii The quadrants where the function is negative.

a Sine function **b** Cosine function

c Tangent function

7 Each of the given points corresponds to the angle θ, measured counterclockwise from the positive x-axis. For each point, find the following ratios:

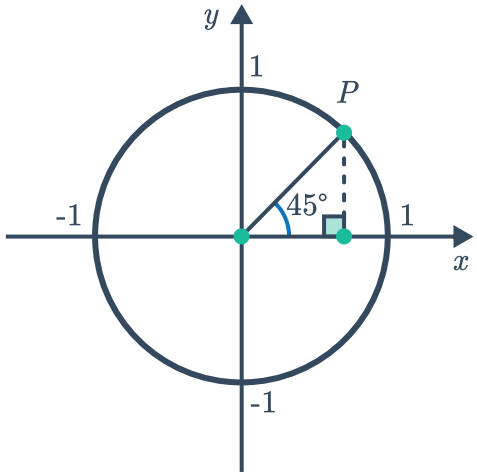
i $\sin\theta$ **ii** $\cos\theta$ **iii** $\tan\theta$ **iv** $\csc\theta$

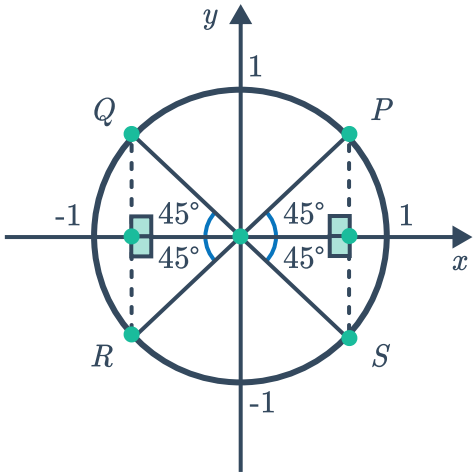
v $\sec\theta$ **vi** $\cot\theta$

a $P\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ **b** $P\left(-\frac{2}{5}, \frac{\sqrt{21}}{5}\right)$ **c** $P\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$

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8 The first diagram shows a unit circle with point $P\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ marked on the circle. Point P represents a rotation of 45° counterclockwise around the origin from the positive x-axis:





a Find the exact values of the following:

i $\sin 45^\circ$ **ii** $\cos 45^\circ$ **iii** $\tan 45^\circ$

b On the second diagram, the coordinate axes shows a 45° angle that has also been marked in the second, third, and fourth quadrants. For each quadrant, find the measure of the relative angle:

i Quadrant 2 **ii** Quadrant 3 **iii** Quadrant 4

c The points Q, R and S mark rotations of point P to the corresponding angles on the unit circle. State the exact coordinates of each point:

i Q **ii** R **iii** S

d Write the following in terms of an equivalent ratio of 45°:

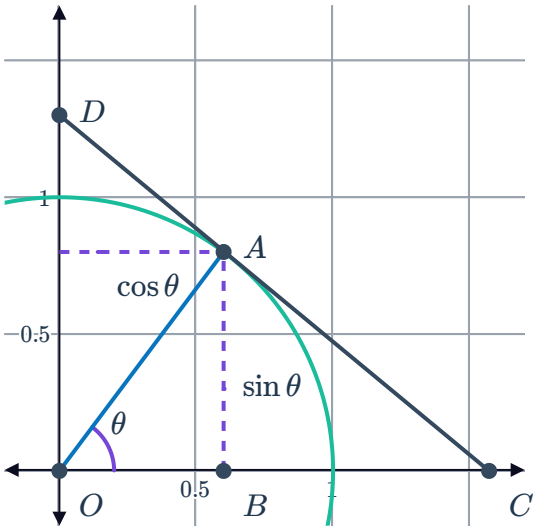
i $\sin 135^\circ$ **ii** $\cos 225^\circ$ **iii** $\tan 315^\circ$ **iv** $\sin(-45)^\circ$

e Now, find the exact value of the following:

i $\sin 135^\circ$ **ii** $\cos 225^\circ$ **iii** $\tan 315^\circ$ **iv** $\sin(-45)^\circ$

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9 In the given diagram, $OA = 1$, and the coordinates of A are $(\cos\theta, \sin\theta)$. $\overline{CD}$ is tangent to the unit circle at A. $\triangle AOB$ is similar to $\triangle COA$ and $\triangle ODA$.



a Show that $AC = \tan\theta$.

b Given that $\sec\theta = \dfrac{1}{\cos\theta}$, show that $OC = \sec\theta$.

c Given that $\csc\theta = \dfrac{1}{\sin\theta}$, show that $OD = \csc\theta$.

d Given that $\cot\theta = \dfrac{1}{\tan\theta}$, show that $AD = \cot\theta$.

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